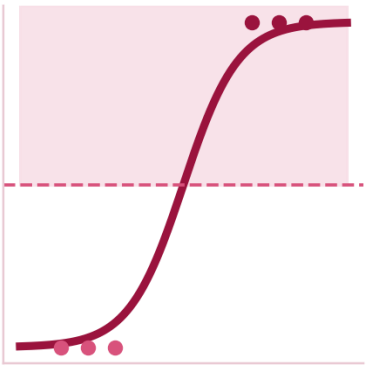
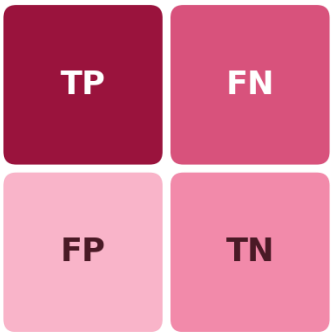
PART A · THEORY

Logistic Regression & Classification Metrics

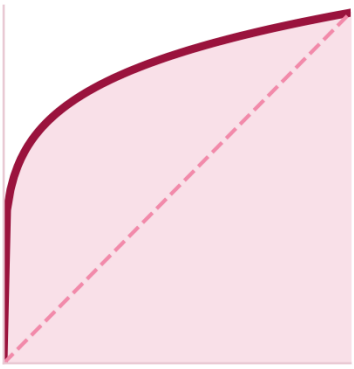
Conceptual Understanding



Logistic Regression



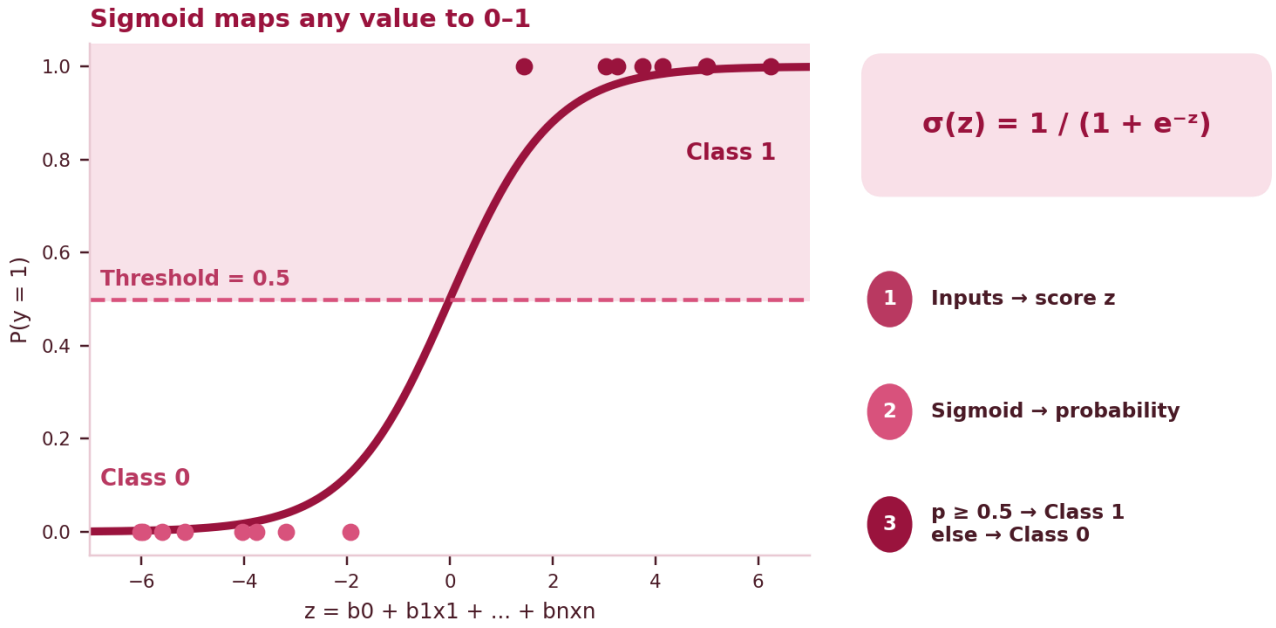
Confusion Matrix



ROC · AUC

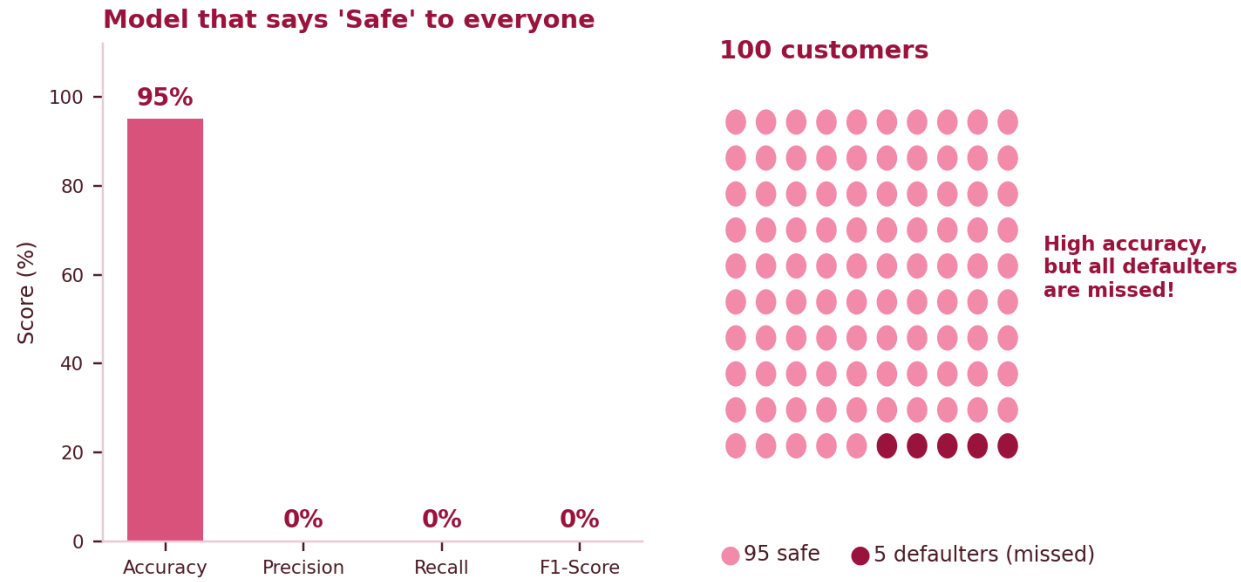
Q1. What is Logistic Regression and why is it suitable for classification?

Logistic Regression is a supervised learning algorithm that predicts the **probability** of a class using the sigmoid function, which squeezes any value into the range 0 to 1. If the probability is 0.5 or more, the record is labelled Class 1, otherwise Class 0. It suits classification because its output is a probability, it is fast, and its coefficients are easy to interpret.



Q2. Explain classification performance metrics and why accuracy alone is insufficient.

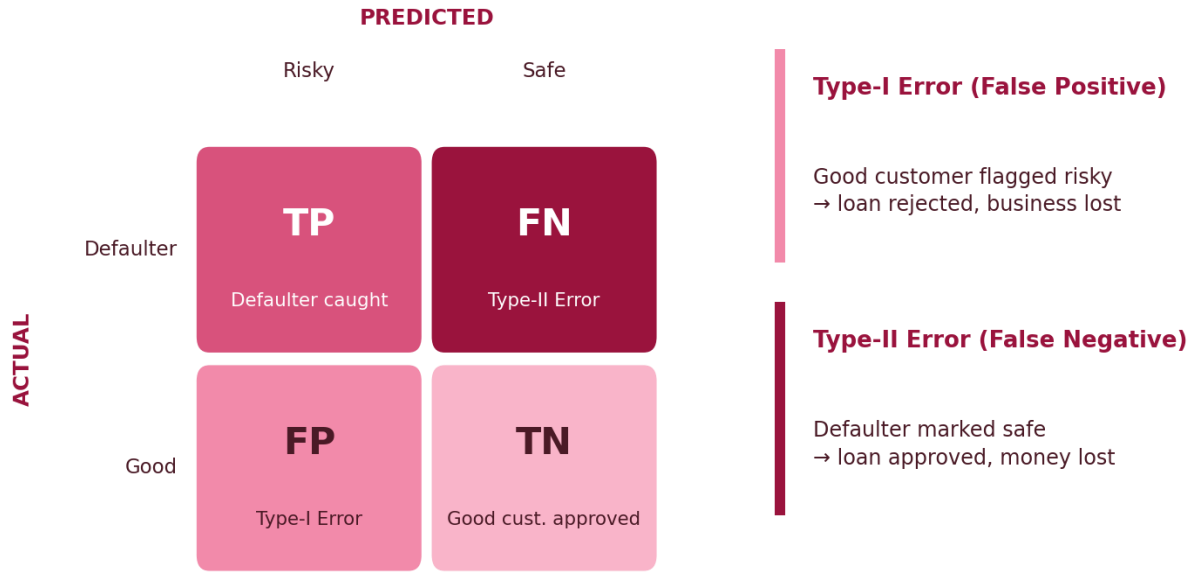
Classification metrics (Accuracy, Precision, Recall, F1-Score, ROC-AUC) compare predicted classes with actual classes using the confusion matrix. **Accuracy alone fails** on imbalanced data: a model that says "Safe" to everyone scores 95% yet catches zero defaulters, so we also need Precision, Recall and F1.



Q3. Define Type-I Error and Type-II Error in the context of risk prediction.

Type-I Error (False Positive): a good customer is wrongly predicted as risky, so a safe loan is rejected and business is lost. **Type-II Error (False Negative):** a real defaulter is wrongly predicted as safe, so the loan is approved and money is lost. In risk prediction, Type-II is usually more costly.

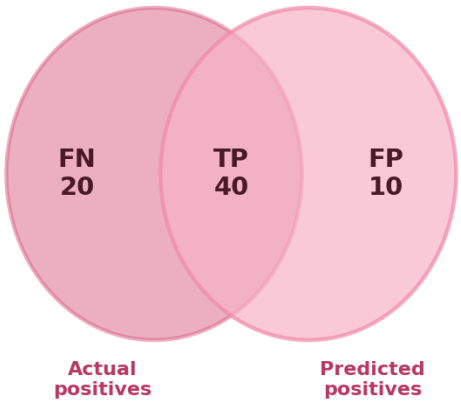
Loan default prediction



Q4. Explain Precision, Recall, F1-Score, TPR and FPR.

Precision = TP/(TP+FP): of all predicted positives, how many are correct. **Recall = TPR** = TP/(TP+FN): of all actual positives, how many are caught. **F1** is the harmonic mean of Precision and Recall. **FPR** = FP/(FP+TN): share of actual negatives wrongly flagged as positive.

TP, FP, FN, TN at a glance



Precision
TP / (TP + FP) **= 0.80**

Recall (TPR)
TP / (TP + FN) **= 0.67**

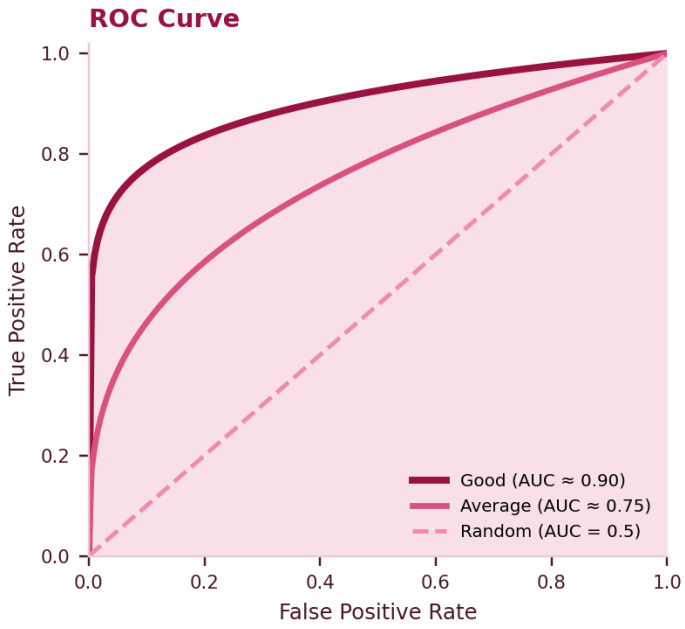
F1-Score
2PR / (P + R) **= 0.73**

FPR
FP / (FP + TN) **= 0.07**

TN = 130

Q5. What is AUC-ROC and how does it help in evaluating classifiers?

ROC curve plots TPR against FPR at every possible threshold, and **AUC** is the area under it (0.5 = random guessing, 1.0 = perfect). It judges a model without depending on one fixed threshold, makes comparing models easy, and helps choose the best cutoff.



How to read AUC

0.9 – 1.0	Excellent
0.8 – 0.9	Good
0.7 – 0.8	Fair
0.5 – 0.7	Weak
0.5	Random

Q6. Why does imbalanced data create problems in classification models?

When one class dominates (e.g. 95% non-default vs 5% default), the model learns to **favour the majority class**, giving high accuracy but poor recall on the minority class that matters most. Fixes include **SMOTE / oversampling, undersampling, class weights**, and judging the model with Recall, F1 or AUC instead of accuracy.

